1. P-values & Confidence Intervals

These are the primary tools for quantifying uncertainty. They transform "guesses" into "evidence."

Technical Core

- **P-value:** The probability of observing test statistics as extreme as the one observed, assuming the Null Hypothesis (H_0) is true. It quantifies the *strength of evidence* against H_0.
    - Formula component (Z-score example):

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$

- **Confidence Interval (CI):** An estimated range of values which is likely to include an unknown population parameter, calculated from a given set of sample data.
    - Formula:

$$\bar{x} \pm Z \frac{\sigma}{\sqrt{n}}$$

    - *Advanced Note:* A 95% CI means if we repeated the experiment 100 times, 95 of the calculated intervals would contain the true population mean. It does **not** mean there is a 95% chance the mean is in *this specific* interval.

Use Cases & Limitations

Feature	Technical Application	Business/Real World Use Case
P-value	Determining feature selection significance in Regression Models.	**Clinical Trials:** Deciding if a new drug's side effects are statistically different from a placebo.
Confidence Interval	Estimating the error margin of a machine learning model's accuracy.	**Polling:** "Candidate A has 48% of the vote ±3%." (Real support is likely between 45-51%).

Critical Limitations (The "Gotchas")

- **P-hacking:** Checking multiple variables until one appears "significant" by chance.

- **Sample Size Sensitivity:** With a large enough sample size (n), even tiny, irrelevant differences become statistically significant ($p < 0.05$). *Always check Effect Size along with P-value.*
- **Misinterpretation:** A high p-value does not prove the Null Hypothesis is true; it only indicates a failure to reject it (insufficient evidence).

The P-value answers one specific question: **"If there were actually NO difference (Null Hypothesis is true), how weird is my data?"**

- **Low P-value:** Data is weird. Something distinctive is happening. (Reject Null).
- **High P-value:** Data is normal/expected. Nothing special is happening. (Fail to reject Null).

P-Value Range	Symbol	Interpretation	Decision
< 0.01	***	**Highly Significant.** extremely unlikely to occur by chance.	**Reject H_0**
< 0.05	*	**Significant.** The standard threshold for "real" results.	**Reject H_0**
0.05 - 0.10	.	**Marginal.** "Trending" towards significance, but not conclusive.	**Usually Fail to Reject H_0**
> 0.10	n.s.	**Not Significant.** Likely just random noise.	**Fail to Reject H_0**

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2. Hypothesis Testing (t-test & ANOVA)

This is the rigorous framework for comparing groups to determine if differences are real or due to random noise.

Technical Core

- **t-test:** Used to compare the means of two groups.
  - *Assumptions:* Normality of data, homogeneity of variance (homoscedasticity), independence of observations.
  - *Types:* Independent (Group A vs B), Paired (Person A Before vs Person A After).
- **ANOVA (Analysis of Variance):** Generalization of the t-test for 3+ groups.
  - It tests if at least one group mean is different by comparing variance **between** groups vs. variance **within** groups.
  - F-Statistic Formula:

$$F = \frac{\text{Between-Group Variability}}{\text{Within-Group Variability}}$$

Use Cases & Limitations

Test Type	Technical Application	Business/Real World Use Case
Independent t-test	Comparing average inference time of Model A vs. Model B.	**Marketing:** Comparing average customer spend of "Email Campaign Users" vs. "Organic Users."
Paired t-test	Evaluating model performance on the *same* dataset before and after hyperparameter tuning.	**HR/Training:** Assessing employee productivity scores *before* and *after* a training workshop.
One-way ANOVA	Comparing error rates across 3 different manufacturing machines.	**Retail:** Comparing average sales across stores in three different regions (North, South, West).

Critical Limitations

- **Normality Assumption:** If data is highly skewed (e.g., income data), t-tests fail. You must switch to non-parametric tests like **Mann-Whitney U** or **Kruskal-Wallis**.

- **The Multiple Comparison Problem:** Using multiple t-tests instead of ANOVA increases Type I error (False Positives). *Correction:* If ANOVA is significant, use **Tukey's HSD** or **Bonferroni correction** for pairwise comparisons.

Component	Description	Python Syntax
Test Name	**One-Way ANOVA** (f_oneway)	scipy.stats.f_oneway
Input Data	3+ independent arrays of numerical data.	f_stat, p = stats.f_oneway(group1, group2, group3)
Assumptions	1. Data is normally distributed. 2. Variances are roughly equal (Homoscedasticity).	stats.shapiro(data) (Normality) stats.levene(data) (Equal Variance)
Result	**P < 0.05:** At least one group differs. **P > 0.05:** All groups are effectively the same.	if p < 0.05: print("Reject Null")

Next Step	If Significant $\rightarrow$ **Post-Hoc Test** (Tukey's HSD) to find *which* pair differs.	statsmodels.stats.multicomp.pairwise_tukeyhsd

Use this step-by-step logic when analyzing multiple groups.

1. **Check Assumptions:**
   - Is the data bell-shaped? (Visual check or Shapiro test).
   - Do groups have similar spread/variance? (Levene test).
   - *If NO:* Use **Kruskal-Wallis Test** (Non-parametric version of ANOVA).
2. **Run ANOVA (f_oneway):**
   - Get F-stat and p-value.
3. **Check P-Value:**
   - **Is $p > 0.05$?** $\rightarrow$ **STOP.** No difference found. Report "No significant difference between groups."
   - **Is $p < 0.05$?** $\rightarrow$ **CONTINUE.** There is a difference, but you don't know where yet.
4. **Run Post-Hoc (Tukey's HSD):**
   - Compare all pairs (A vs B, B vs C, A vs C).
   - Identify exactly which specific group is driving the difference.

Python Example Code (Copy-Paste Ready)

Python

```
from scipy import stats

import statsmodels.stats.multicomp as mc


# 1. The Data (e.g., User scores from 3 different UI designs)

design_a = [85, 88, 90, 82, 86]

design_b = [75, 78, 72, 79, 74]

design_c = [84, 86, 88, 85, 87]
```

```
# 2. Run ANOVA
f_stat, p_val = stats.f_oneway(design_a, design_b, design_c)

print(f"P-value: {p_val:.5f}")

# 3. Decision Logic
if p_val < 0.05:
    print("Significant difference found! Running Post-Hoc...")

    # Prepare data for Tukey's HSD (requires stacked data)
    data = design_a + design_b + design_c
    labels = ['A']*5 + ['B']*5 + ['C']*5

    # Run Tukey's HSD
    tukey = mc.pairwise_tukeyhsd(endog=data, groups=labels, alpha=0.05)
    print(tukey)
else:
    print("No significant difference found.")
```

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3. A/B Testing Fundamentals (The Application)

A/B testing is the gold standard for causal inference in product and business analytics. It is essentially a randomized controlled experiment.

Technical Core

- **Randomization:** Users are randomly assigned to Control (A) or Treatment (B) to eliminate confounding bias.

- **Power Analysis:** Calculation performed *before* the test to determine the minimum sample size (n) required to detect a specific effect size (Minimum Detectable Effect - MDE) with a given power (usually 80%) and significance level ($\alpha = 0.05$).
  - $$n \propto \frac{\sigma^2}{\delta^2}$$
    (where δ is the difference you want to detect).
- **Metric Selection:**
  - *Guardrail Metrics:* Metrics that shouldn't degrade (e.g., Page Load Time).
  - *North Star Metrics:* The primary success metric (e.g., Conversion Rate).

Use Cases & Limitations

Component	Technical Application	Business/Real World Use Case
Binary Metric (0/1)	Analyzing click-through rates (Bernoulli distribution).	**UI/UX:** Testing if a "Red Button" gets more clicks than a "Blue Button."
Continuous Metric	Analyzing session duration or revenue per user.	**Pricing:** Testing if a $9.99 subscription price yields higher total revenue than $12.99.

Critical Limitations

- **Peeking Problem:** Checking results continuously and stopping when significant leads to high false positive rates. *Solution:* Fix the sample size in advance or use **Sequential Testing** methods.
- **Novelty Effect:** Users might click a feature just because it's new, not because it's better. The effect fades over time.
- **Network Effects:** In social networks (like Facebook/LinkedIn), Treatment users might affect Control users (Interference), violating the independence assumption (SUTVA).

Summary of Limitations & Advanced Alternatives

Concept	Limitation	Advanced Alternative / Fix

P-values	Easy to hack; ignores effect size.	**Bayesian Inference:** Calculate the probability the hypothesis is true given the data $P(H
t-test	Assumes Normal distribution.	**Bootstrapping:** Resampling data to estimate distribution empirically.
A/B Testing	Expensive; takes time to reach significance.	**Multi-Armed Bandits:** Algorithms (Thompson Sampling) that dynamically route traffic to the winning variation *during* the test to maximize reward.

Statistical Test	Theory (When to use)	Tech Scenario (Example)	Python Function & Syntax	Interpretation of Result
Independent T-Test	Compares the means of two **separate, unrelated** groups. *(Parametric)*	**A/B Testing:** Comparing Average Revenue Per User (ARPU) between Group A (Red Button) and Group B (Blue Button).	from scipy import stats stat, p = stats.ttest_ind(group_a, group_b)	If $p < 0.05$: The difference in revenue is **real**, not just random chance.

Paired T-Test	Compares means of the **same group** at different times. *(Dependent)*	**Performance Tuning:** Measuring server response time **before** vs. **after** a code deployment on the *same* specific servers.	from scipy import stats stat, p = stats.ttest_rel(before, after)	If $p < 0.05$: The deployment caused a statistically significant change in speed.
One-Way ANOVA	Compares means across **3 or more** groups to see if *any* differ.	**Feature Analysis:** Comparing average "Time on App" across 3 different user segments: (1) Gen Z, (2) Millennials, (3) Boomers.	from scipy import stats stat, p = stats.f_oneway(group1, group2, group3)	If $p < 0.05$: At least one generation behaves differently. (Run post-hoc tests to find which).
P-Value	Probability that the observed result occurred by random chance assuming H_0 is true.	"Is this conversion lift real, or noise?"	Returned as the second value in all functions above (variable p).	**Low P-Value (< 0.05):** Reject Null Hypothesis (Result is Significant).

				High P-Value (> 0.05): Result is likely noise.
Confidence Interval	A range (e.g., 95%) where the *true* population mean likely sits.	Estimating the "true" average load time for all users based on a sample of 1000 users.	`stats.t.interval(alpha=0.95, df=len(data)-1, loc=mean, scale=sem)`	"We are 95% confident the true load time is between 200ms and 210ms."

Phase	Action	Technical Note
1. Hypothesis	Define H_0 (No difference) and H_1 (Difference exists).	"The new checkout flow (H_1) will increase conversion rate compared to old flow (H_0)."
2. Sampling	Calculate needed sample size (Power Analysis).	Use `statsmodels.stats.power` to ensure you don't stop the test too early.
3. Testing	Collect data and run the **Independent T-Test** (for means) or **Z-Test** (for proportions).	Do not peek at P-values daily; wait for sample size to be met.

4. Decision	Check if $p < 0.05$ (Significance) AND check effect size (Business impact).	A result can be statistically significant ($p=0.04$) but not worth the cost of engineering (0.01% lift).